

VALLURI DINESH RAM

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AI/ML Engineer | Agentic Systems | Generative AI

SKILLS

Languages & Database : Python, C, HTML, CSS, SQL, MongoDB, FAISS, ChromaDB, Pinecone Vector Database
Frameworks & Stack : ScikitLearn, Tensorflow, Keras, Langchain, LangGraph, Streamlit, LLM, OpenCV, DSA
Generative AI, Agentic AI, Prompt Engineering, **Machine Learning**, Deep Learning
Cloud & Workflow : AWS, GitHub, Git, MLOps, Django, Flask, FastAPI, CI/CD, GitHub Actions, Docker

EXPERIENCE

Junior AIML/Computer Vision Engineer , Tractrix Opto Dynamics ([website](#)) Feb 2025 - Present

- Architected a Self-Reflective RAG(Retrieval Augmented Generation) system with an agentic, stateful graph pipeline featuring adaptive retrieval gating, LLM-based document relevance scoring, and multi-stage conditional routing, enabling accurate, context grounded Q&A over enterprise knowledge bases. Automated hallucination mitigation loop with post generation grounding verification that classifies answers as supported or unsupported, and triggers autonomous revision cycles before delivery to end users. Graph based AI orchestration workflows with conditional routing, retry controls, recursive validation, and self corrective paths aligned with AI system.
- Led optimization of real time video pipeline (GStreamer) for edge AI devices, reduced end to end inference latency by 35% and increased throughput from 18 FPS to 30 FPS, improved frame processing throughput by 2× and reduced CPU utilization by 40%. Designed and shipped modular Flask microservices for camera ingestion, inference, and various features. Engineered and built Recording, transfer of videos, Client Server architecture, database state management, stabilization systems, workflows, data annotations. Extensive working on Embedded/Edge AI devices.

Artificial Intelligence Intern , Smart Prepper ([website](#)) Aug 2024 - Nov 2024

- Developed an end to end Flashcard Generator that accepts text, topic, and multi-page PDF input (PDFs stored on AWS S3) and produces Q&A flashcards using LLM prompts and RAG retrieval for context automated generation of 1,000+ flashcards per hour. Integrated secure PDF upload and multi-page text extraction workflow (AWS S3 buckets), and instrumented analytics to track usage, achieving more than 80% successful conversion.
- Built a Mind Map Generator (Django and OpenAI) that converts long, unstructured notes into Markdown mind maps; cut manual summarization time by 70% for user trials targeted at UPSC aspirants.
- Implemented a scalable pipeline using vector embeddings and Pinecone for semantic retrieval to support quiz and MCQ generation, improved MCQ relevance by 25% in user and internal test.

Lead Technology, GDSC(Google Developer Students Club) and NLP Stud, IIIT H Aug 2023 - Aug 2024

- As Lead, conducted sessions on machine learning, deep learning usage of TensorFlow, Colab, Cloud Study Jams. As NLP program, developed speaker recognition system with 97% accuracy using CNN architecture. ([Link](#))

PROJECTS

MeetingMind: AI Powered Meeting Orchestration Agent: Built an end-to-end agentic AI workflow using LangGraph that processes meeting transcripts through a multi-agent architecture: (Understanding, Extraction, Reflection, Routing, Execution) and using structured outputs via Pydantic schemas and monitored all traces using LangSmith observability. Implemented Human in the Loop (HITL) system allowing users to review, approve, or reject actions before MCP tool execution. **Integrated 4 MCP (Model Context Protocol)** servers which are Gmail, Google Calendar, Jira, and Slack, enabling auto task execution like sending emails, scheduling meetings, and creating tickets. Built backend with FastAPI and database via SQLAlchemy ORM with SQLite. Developed a Streamlit dashboard for user interaction: transcript upload, action review, one click approval/rejection, execution monitoring, and audit trail viewing. ([GitHub](#))

Advanced Multi Agent AI Content: Designed 5 agent stateful orchestration pipeline (Router, Research, Orchestrator, parallel Workers, Reducer) using LangGraph; Classifier routes across 3 execution modes (closed/hybrid/open). Integrated Tavily tool for real time web research with evidence and citation; LangSmith observability captures all traces, FastAPI for backend. Deployed on AWS EC2 via Docker, GitHub Actions CI/CD pipeline with auto deployment. ([GitHub](#))

EMMA-Local: Healthcare voice reception agent: Built with safety guardrails, intent router, self-corrective verifier; hybrid RAG (BM25, dense vectors, RRF, cross-encoder reranking, HyDE), Whisper voice pipeline, FastAPI.

ACHIEVEMENTS & EDUCATION

Paper Publication: ([Link](#)) , MIT iQuHACK 2023: Microsoft's challenge, AMAZON ML CHALLENGE, NPTEL: Elite, National Hackathons, CISCO CCNA, IBM, Udemy certifiks. **B.Tech** at HITAM , **CSE** , CGPA : **9.01** , 2021 - 2025